UCS503 – SOFTWARE ENGINEERING

MeterIQ

Explainable Smart Meter Analytics Platform

Prototype Report

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1 Introduction

1.1 Background

Electricity distribution networks generate large amounts of smart meter data at regular time intervals. Analysing this data can help identify unusual consumption behaviour and support better understanding of electricity usage.

MeterIQ is an explainable smart meter analytics platform designed to analyse smart meter consumption data. The system focuses on two major analytical functionalities: anomaly or electricity-theft-risk detection and short-term load forecasting.

The platform is being developed using smart meter data from Mathura and Bareilly. Instead of directly declaring a meter as a theft case, MeterIQ identifies unusual patterns and provides information that can support further human investigation.

1.2 Problem Statement

Traditional analysis of smart meter data can make it difficult to identify unusual consumption patterns efficiently. Large datasets also make manual inspection time-consuming.

The proposed system aims to provide an integrated platform where smart meter data can be processed, analysed and visualised through a dashboard. The system also aims to provide explainable information along with detected anomalies so that the results can be understood by users.

1.3 Objectives

The main objectives of MeterIQ are:

- Analyse smart meter consumption data.
- Detect unusual consumption patterns and generate anomaly or theft-risk indicators.
- Forecast short-term electricity load.
- Provide explanations for detected unusual behaviour.
- Present analytical results through an interactive dashboard.
- Support human investigation rather than automatically accusing a consumer of electricity theft.

2 System Design

2.1 System Overview

The MeterIQ system follows a data-driven workflow. Smart meter data is first prepared and transformed into suitable analytical features. The prepared data is then used by the anomaly detection and forecasting components. The resulting information is presented through the dashboard.

The overall workflow is:

Smart Meter Data → Data Preparation → Feature Engineering → Analytics → Dashboard

The analytics layer consists of two primary modules:

1. Anomaly / electricity-theft-risk detection.
2. Short-term load forecasting.

2.2 Use Case Diagram

The Use Case Diagram describes the major interactions between the Smart-Meter Data Source, User and the MeterIQ system.

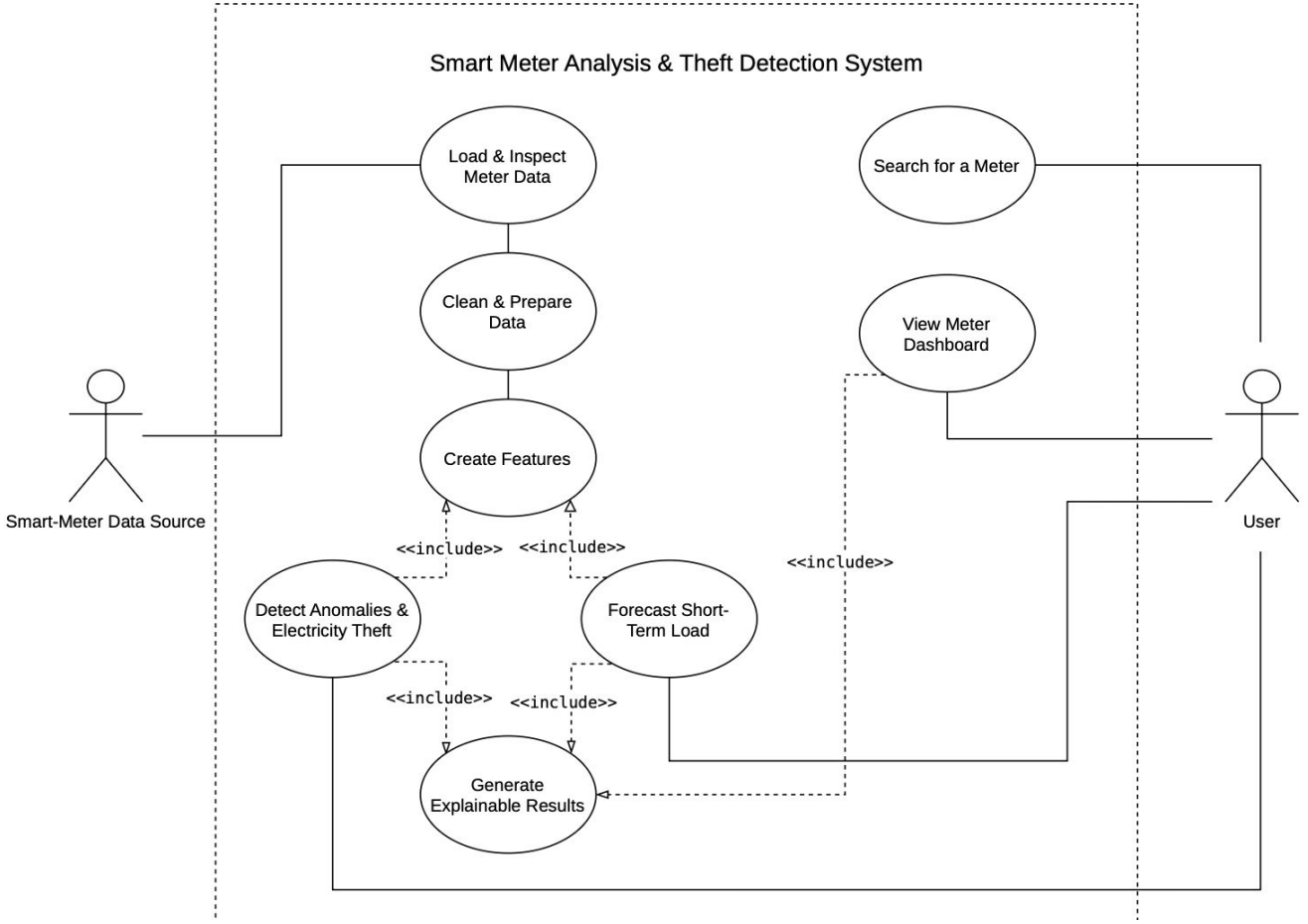


Figure 2.1: MeterIQ Use Case Diagram

3 System Modelling

This chapter presents the system models that define how MeterIQ components, users and data stores interact. The diagrams progress from a high-level data flow to detailed processing, the data model and the end-to-end activity workflow.

3.1 Data Flow Diagram – Level 0

The Level 0 Data Flow Diagram provides a high-level representation of the MeterIQ system and shows the interaction between the system and its external entities.

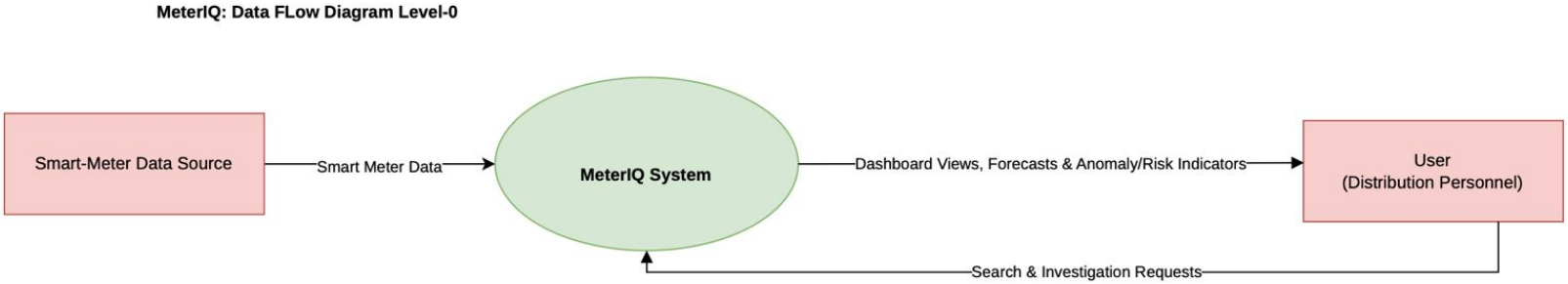


Figure 3.1: MeterIQ DFD Level 0

3.2 Data Flow Diagram – Level 1

The Level 1 DFD provides a detailed representation of the major processes involved in smart meter data analysis.

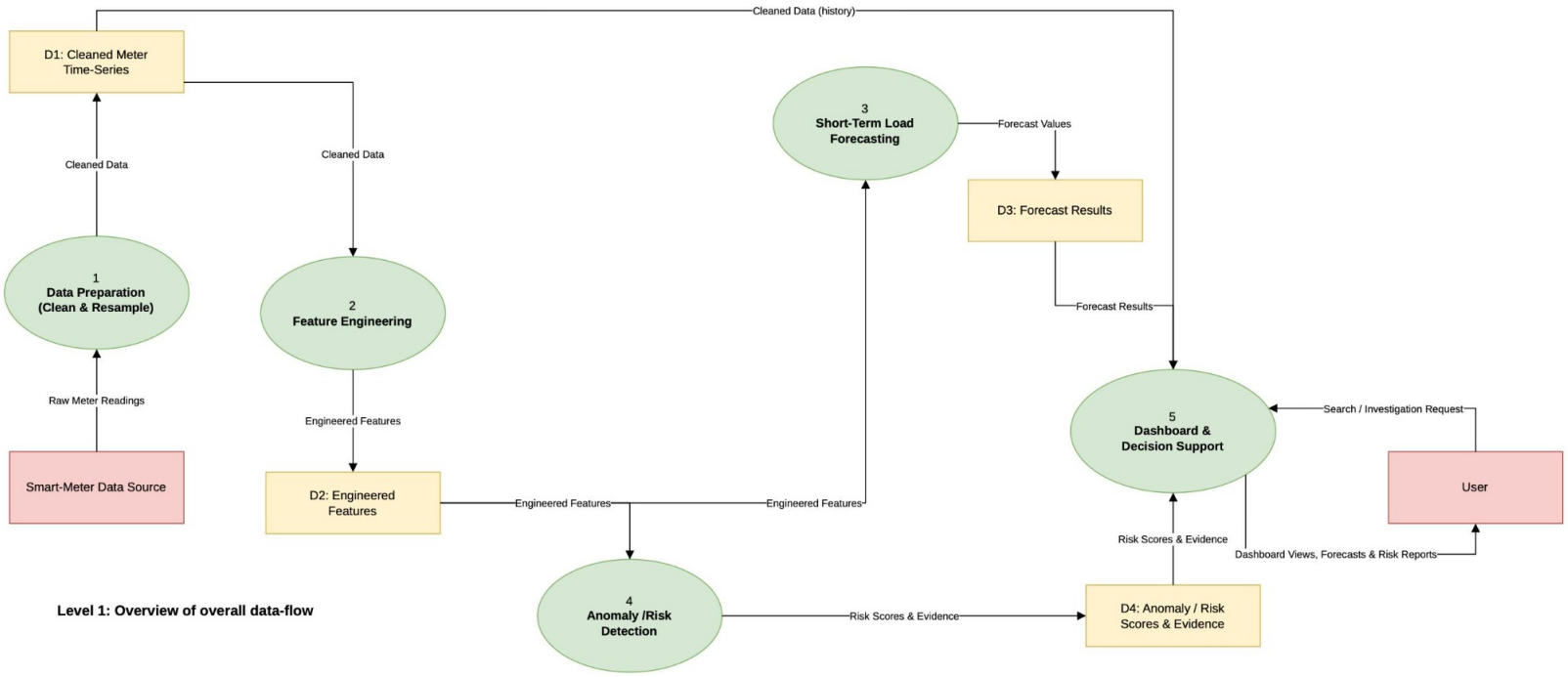


Figure 3.2: MeterIQ DFD Level 1

3.3 Data Flow Diagram – Level 2

The Level 2 DFD further decomposes the anomaly and theft-risk detection process and represents the flow of information between its components.

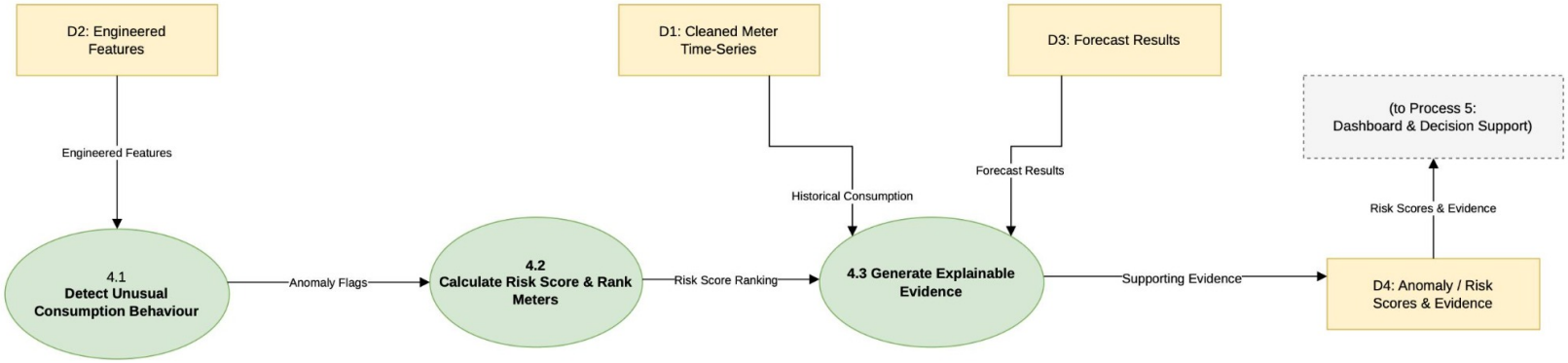


Figure 3.3: MeterIQ DFD Level 2

3.4 Entity Relationship Diagram

The Entity Relationship Diagram represents the major data entities and their relationships used in the MeterIQ system.

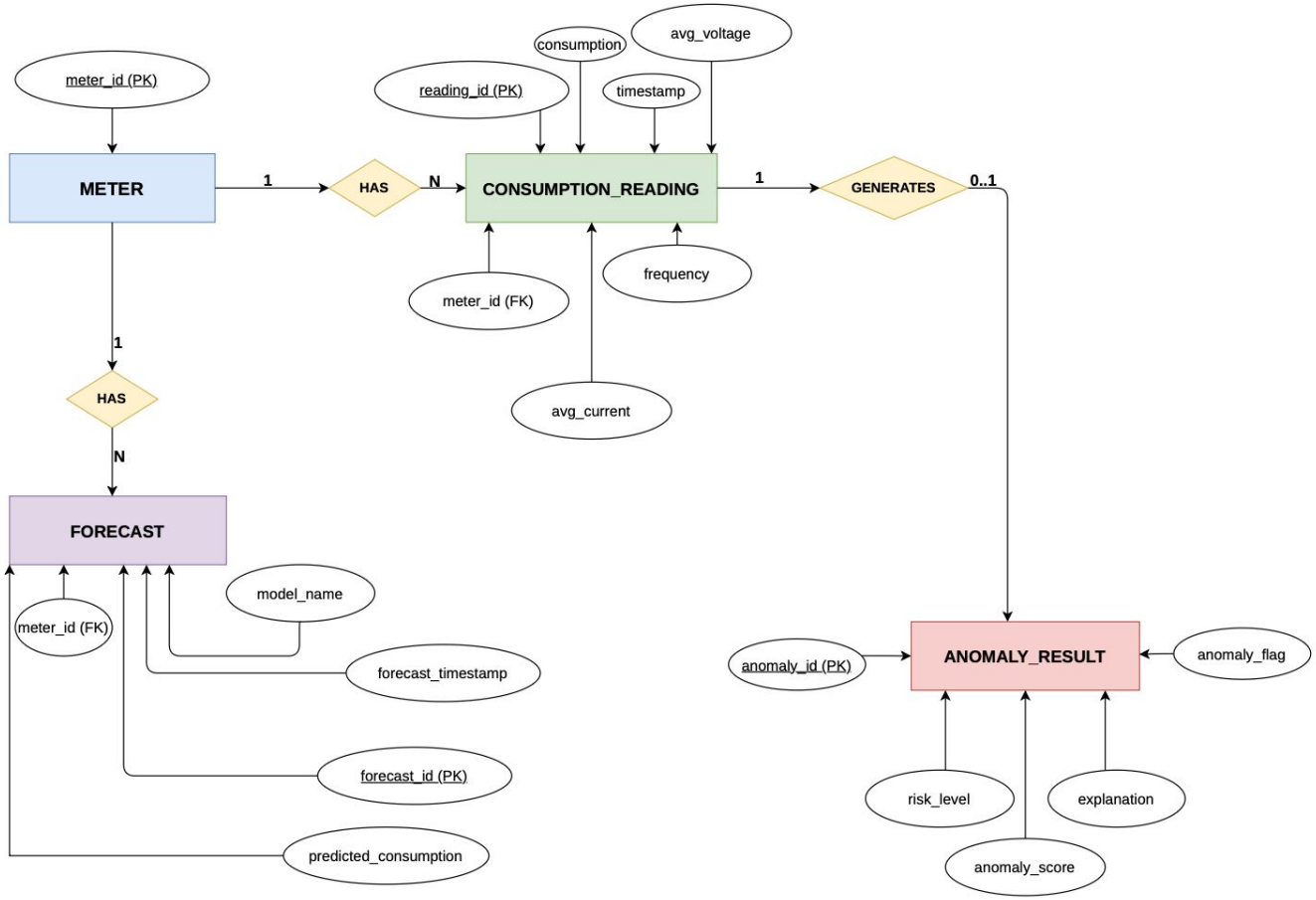


Figure 3.4: MeterIQ Entity Relationship Diagram

3.5 Activity Diagram

The Activity Diagram represents the sequence of activities performed during the MeterIQ analysis workflow, from receiving meter data to generating analytical results.

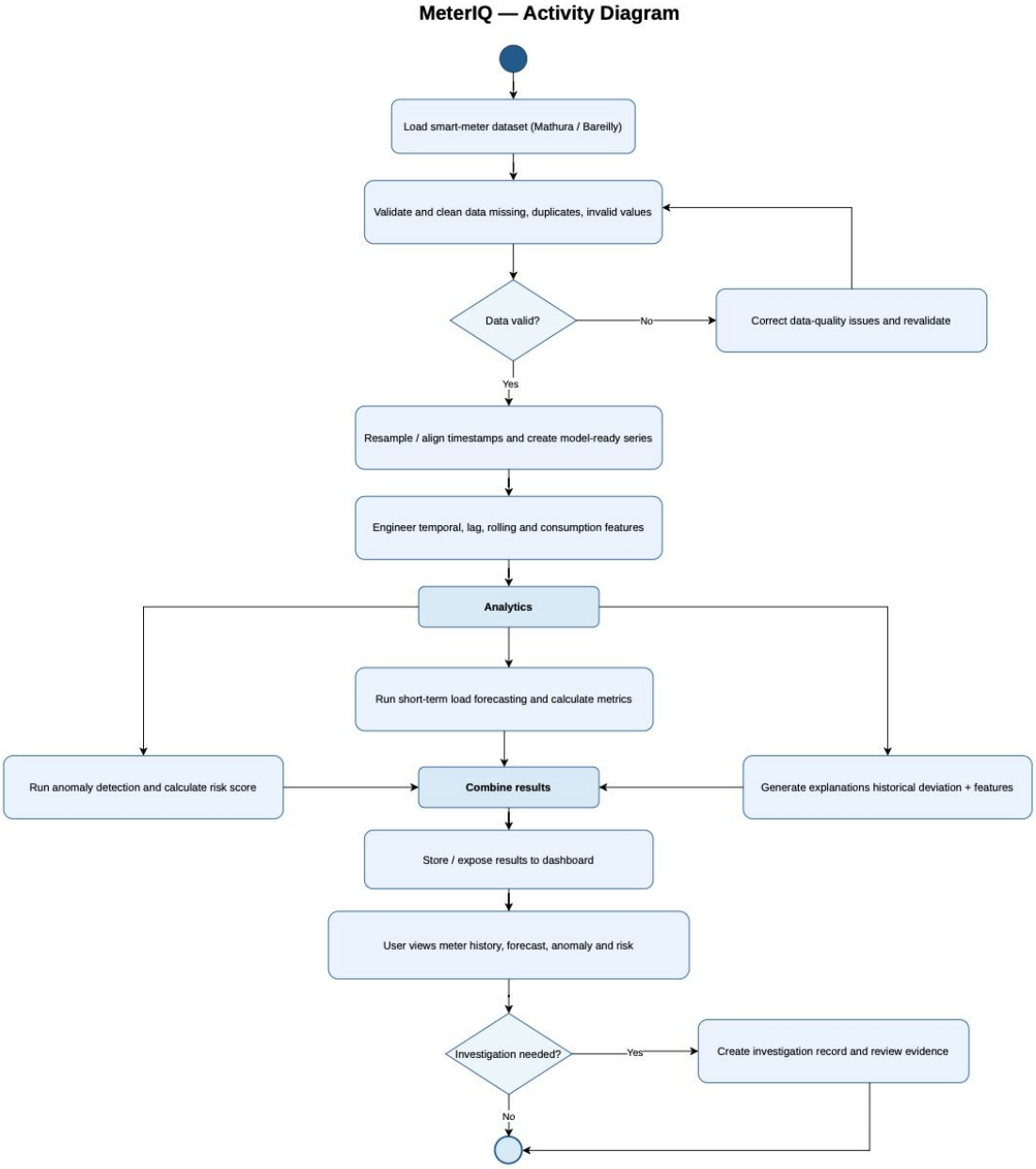


Figure 3.5: MeterIQ Activity Diagram

4 Prototype Implementation

4.1 Data Preparation

The smart meter data is prepared before performing analytical operations. The preprocessing stage includes handling the available meter readings, organising the time information and preparing the data for further analysis.

Feature engineering is then performed to generate useful temporal and consumption-related features. These features are used by the analytical modules.

4.2 Anomaly and Risk Detection

The anomaly detection component is intended to identify unusual patterns in smart meter consumption data.

The system does not directly classify a consumer as committing electricity theft. Instead, it produces an anomaly or risk indicator based on unusual behaviour. The result can then be examined by an authorised human user.

The explainability component is intended to provide information about why a reading or pattern has been identified as unusual.

4.3 Short-Term Load Forecasting

The forecasting component is designed to estimate electricity load for a short-term horizon. A same-hour-last-week baseline is used as an initial reference for evaluating forecasting performance.

The forecasting workflow uses chronological data separation so that future information is not used during the training process.

4.4 Dashboard

A web-based dashboard is being developed to provide a user-friendly interface for MeterIQ. The dashboard is intended to allow users to search and inspect meters, view consumption patterns, examine anomaly or risk information and view load forecasts.

The dashboard provides a single interface for presenting the results of the analytical pipeline.

4.5 Frontend Prototype

The MeterIQ frontend provides an interactive interface for smart meter monitoring, analysis and decision support. The prototype includes a login interface, an operations dashboard and a short-term load forecasting interface.

4.5.1 Login Interface

The login interface provides access to the MeterIQ platform and supports different prototype roles, including Supervisor and Data Analyst.

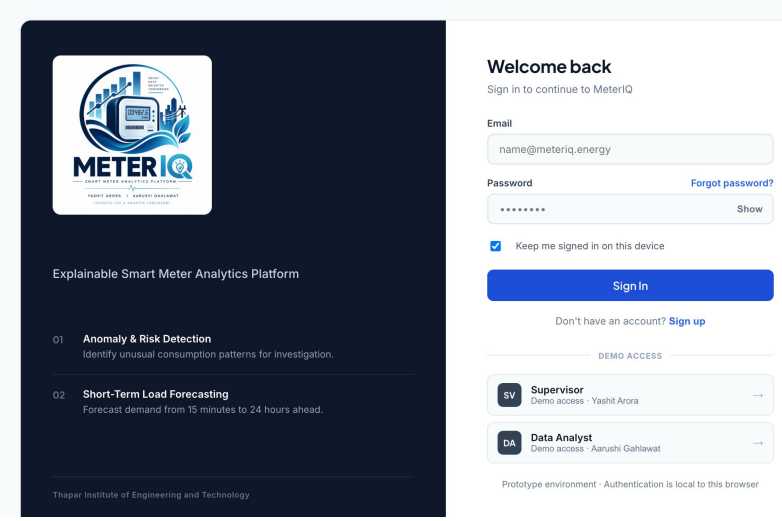


Figure 4.1: MeterIQ Login Interface

4.5.2 Operations Dashboard

The Operations Dashboard provides an overview of smart meter activity. It presents total smart meters, total consumption, detected anomalies, high-risk meters and active regions. It also provides consumption trends and anomaly-risk distribution for operational decision support.

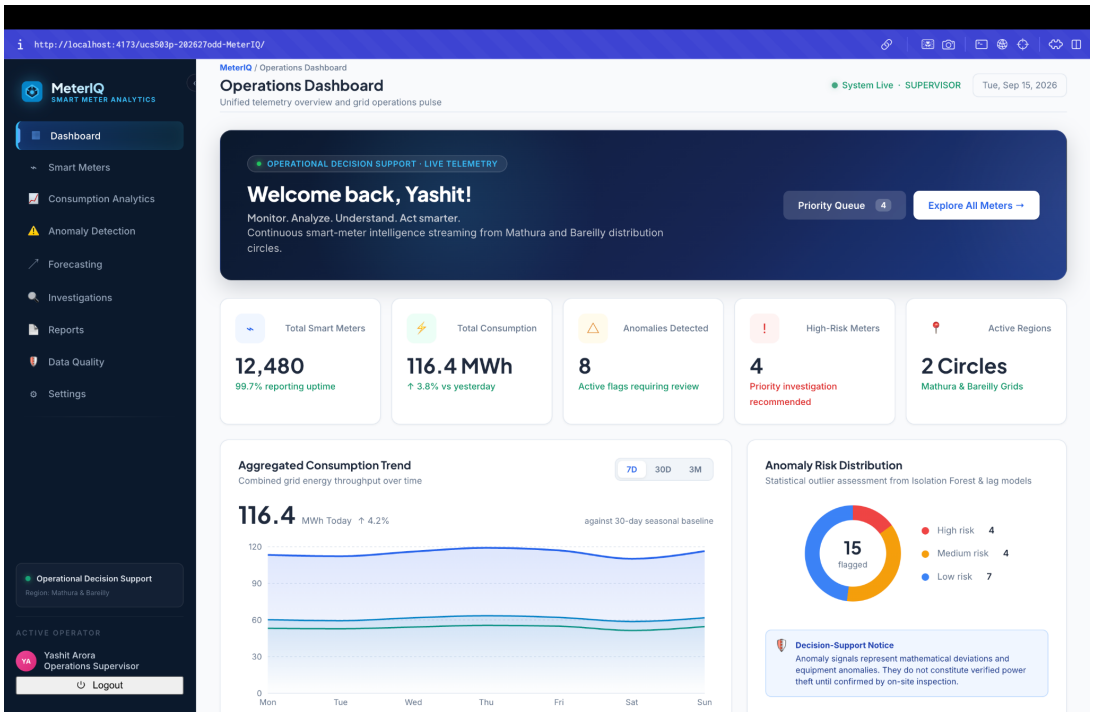


Figure 4.2: MeterIQ Operations Dashboard

4.5.3 Short-Term Load Forecasting Interface

The forecasting interface allows a smart meter and machine learning model to be selected along with the forecast horizon. It presents forecasting evaluation metrics and the predicted load profile for the selected meter.

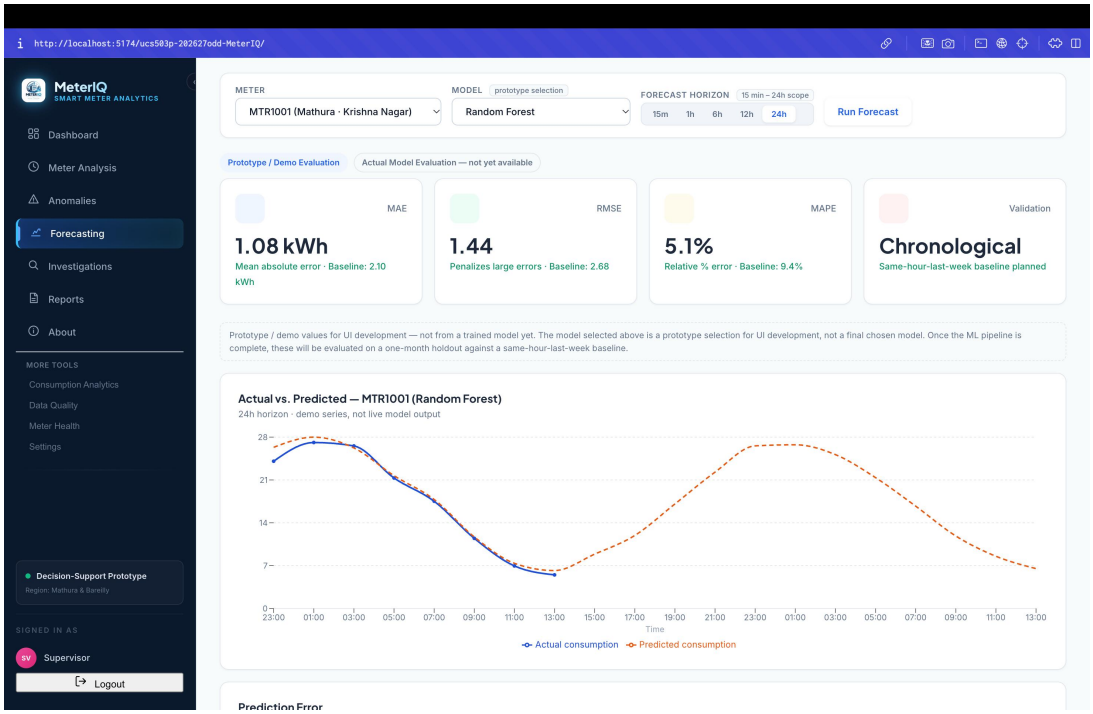


Figure 4.3: MeterIQ Short-Term Load Forecasting Interface

5 Current Progress and Future Work

The current prototype development includes the following completed or developed components:

- Problem definition and system requirements.
- Use Case Diagram.
- DFD Level 0, Level 1 and Level 2.
- Entity Relationship Diagram.
- Activity Diagram.
- Exploratory Data Analysis.
- Data preprocessing.
- Feature engineering.
- Initial dashboard development.
- System workflow and prototype design.

5.1 Prototype Status

The prototype currently demonstrates the planned system structure and dashboard workflow. The final analytical results and model evaluation will be added after completion and integration of the machine learning modules.

The current report represents the mid-semester prototype stage of MeterIQ rather than the final project evaluation.

5.2 Future Work

The planned future work for MeterIQ includes:

- Complete the anomaly and electricity-theft-risk detection module.
- Complete and evaluate the short-term load forecasting module.
- Integrate the analytical models with the dashboard.
- Add explainable anomaly information to the user interface.
- Perform complete system testing.
- Evaluate the system using appropriate performance metrics.
- Improve the dashboard based on testing and user feedback.

Bibliography

[1] Smart meter consumption data used for the MeterIQ project, including data from Mathura and Bareilly.

[2] UCS503 Software Engineering course material, Thapar Institute of Engineering and Technology.

[3] MeterIQ project repository:
https://github.com/aarushi2006/ucs503p-202627odd-MeterIQ